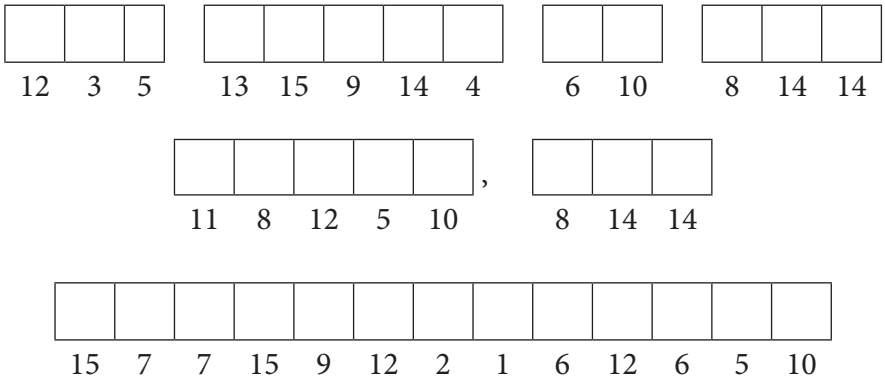
Name: _____

WORKSHEET

Addition and subtraction of matrices

We are all inventors, each sailing out on a voyage of discovery, each guided by a private chart, of which there is no duplicate.



Ralph Waldo Emerson

Questions

Given $\mathbf{A} = \begin{bmatrix} 7 & -3 & 2 \\ 5 & 0 & 6 \\ 1 & -1 & -4 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 9 & 6 \\ 0 & 12 \end{bmatrix}$, $\mathbf{C} = \begin{bmatrix} -8 & 4 & -2 \\ 7 & 1 & 10 \\ -6 & 5 & 9 \end{bmatrix}$,

$\mathbf{D} = \begin{bmatrix} 10 & -8 \\ 2 & -6 \end{bmatrix}$, $\mathbf{E} = \begin{bmatrix} 5 & 1 \\ -3 & 7 \end{bmatrix}$, $\mathbf{F} = \begin{bmatrix} 3 & 3 & -7 \\ 8 & 2 & 0 \\ -6 & -1 & 4 \end{bmatrix}$

Calculate the following if possible:

1 $\mathbf{B} + \mathbf{E}$

2 $\mathbf{A} + \mathbf{F}$

3 $\mathbf{D} + \mathbf{B}$

4 $\mathbf{C} + \mathbf{F}$

5 $\mathbf{A} + \mathbf{E}$

6 $\mathbf{D} - \mathbf{E}$

7 $\mathbf{C} - \mathbf{A}$

8 $\mathbf{A} - \mathbf{C}$

9 $2\mathbf{E}$

10 $\mathbf{A} + 3\mathbf{C}$

11 $2\mathbf{D} - \mathbf{B}$

12 $\mathbf{A} + \mathbf{C} - \mathbf{F}$

13 $-\frac{2}{3}\mathbf{B}$

14 $4\mathbf{C} - 3\mathbf{F}$

15 $\mathbf{E} + \frac{1}{2}\mathbf{D} + \frac{1}{3}\mathbf{B}$

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Solutions

A $\begin{bmatrix} 15 & -7 & 4 \\ -2 & -1 & -4 \\ 7 & -6 & -13 \end{bmatrix}$

D $\begin{bmatrix} -5 & 7 & -9 \\ 15 & 3 & 10 \\ -12 & 4 & 13 \end{bmatrix}$

E Not possible because **A** and **E** are of different orders.

G $\begin{bmatrix} 11 & -22 \\ 4 & -24 \end{bmatrix}$

H $\begin{bmatrix} 19 & -2 \\ 2 & 6 \end{bmatrix}$

I $\begin{bmatrix} 5 & -9 \\ 5 & -13 \end{bmatrix}$

L $\begin{bmatrix} -41 & 7 & 13 \\ 4 & -2 & 40 \\ -6 & 23 & 24 \end{bmatrix}$

N $\begin{bmatrix} 14 & 7 \\ -3 & 19 \end{bmatrix}$

O $\begin{bmatrix} 13 & -1 \\ -2 & 8 \end{bmatrix}$

P $\begin{bmatrix} -15 & 7 & -4 \\ 2 & 1 & 4 \\ -7 & 6 & 13 \end{bmatrix}$

R $\begin{bmatrix} 10 & 2 \\ -6 & 14 \end{bmatrix}$

S $\begin{bmatrix} -17 & 9 & -4 \\ 26 & 3 & 36 \\ -17 & 14 & 23 \end{bmatrix}$

T $\begin{bmatrix} -4 & -2 & 7 \\ 4 & -1 & 16 \\ 1 & 5 & 1 \end{bmatrix}$

U $\begin{bmatrix} 10 & 0 & -5 \\ 13 & 2 & 6 \\ -5 & -2 & 0 \end{bmatrix}$

W $\begin{bmatrix} -6 & -4 \\ 0 & -8 \end{bmatrix}$

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